Report

Overall Observations Insights

- The pass rates of the models across tasks are broadly consistent with their overall average pass rates (see figure 1). This suggests that the benchmark is reasonably robust: models that do well overall also tend to do well on each task variant, and vice versa.
- Task 2 is generally harder than Task 1 (see figure 1). Removing the scale seems to make it difficult for the LLMs to understand movement.

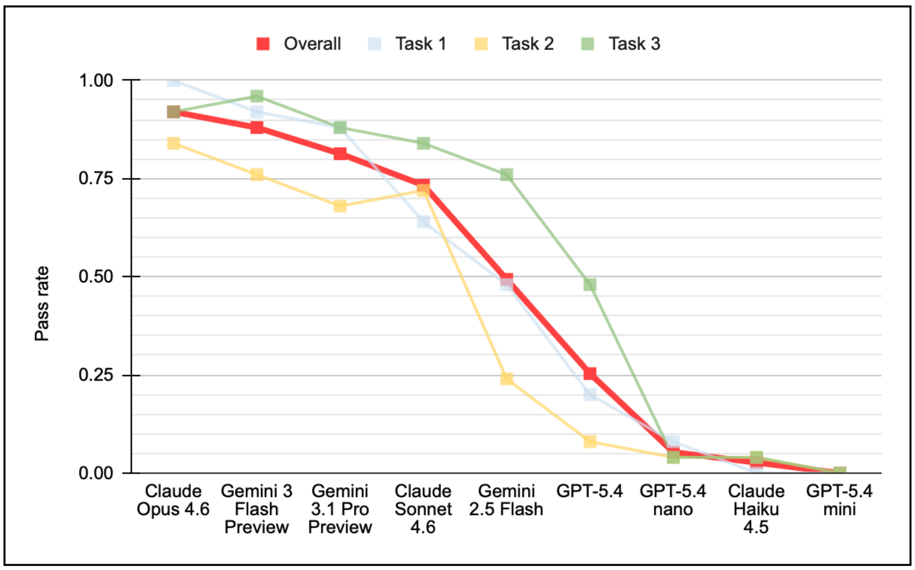

Figure 1

Overall Pass rate	
Claude Opus 4.6	0.92
Gemini 3 Flash Preview	0.88
Gemini 3.1 Pro Preview	0.81
Claude Sonnet 4.6	0.73
Gemini 2.5 Flash	0.49
GPT-5.4	0.25
GPT-5.4 nano	0.05
Claude Haiku 4.5	0.03
GPT-5.4 mini	0.00

Table 1

- While designing the benchmark, the prompt was iteratively refined using feedback from Gemini 3 Flash Preview. This probably may have led to the final benchmark tasks overfitting to that particular model i.e. Gemini 3 Flash Preview may be getting a higher score than it would have on a prompt designed in a more model-agnostic way.

- The same benchmark was also run on a text-only LLM (without image input). In several cases, it was able to come up with the right final sequence just by guessing what each image represents (without any knowledge of the actual image)

> **Deepseek v3.2**:
>
> - reasoning:"Since no images were provided, I will assume a generic case. Typically in stickman puzzle games, actions are: A = jump, B = crouch/duck, C = move sideways. The GOAL says 'avoid red', implying the stickman must not touch red obstacles. Without seeing the images, the shortest safe sequence to avoid red might involve jumping over a low obstacle (A) and then moving sideways to avoid a subsequent threat (C). Assuming one jump is enough, sequence 'AC' could work. But to match a common minimal pattern for 'avoid red' with two actions, I'll propose 'AC'. If the environment requires only a single action, like just jumping, then 'A' would be shortest, but 'avoid red' often implies multiple hazards. Since the instructions ask for the shortest sequence given the GOAL image, and the image isn't shown, I'll assume a plausible minimal sequence based on typical puzzles: 'AC' (jump, then move sideways)."
> - sequence:"AC"

- There is a possibility that even the multi-modal LLMs might have some level of knowledge as to what these images are, before looking at them. This may indicate that they are **not purely learning the conventions from the images**.
- As shown in Figure 2, models that have higher response times tend to perform better on the benchmark. One plausible explanation is that these models allocate more computation or internal deliberation per query, hence leading to increased accuracy.

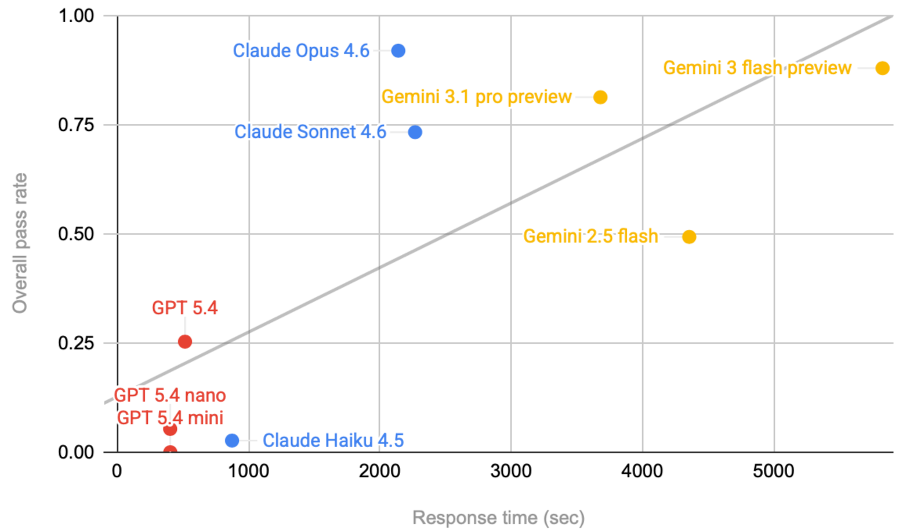

Figure 2

Future Directions:

- Since text models were able to guess the configuration of the task, I wonder how it will be if we make the task a little more harder. (Hard enough that text models cannot easily guess the configuration)
- I want to find out the result if we do not overfit the prompt to any particular model. But how can we adjust the model difficulty without overfitting to any specific model?
- I wonder how the GPT and claude models will perform if we make them reason more by inputting a mandatory reasoning budget in the prompt or use other reliable methods.

Appendix:

Task 1 Pass rate	
Claude Opus 4.6	1
Gemini 3 Flash Preview	0.92
Gemini 3.1 Pro Preview	0.88
Claude Sonnet 4.6	0.64
Gemini 2.5 Flash	0.48
GPT-5.4	0.2
GPT-5.4 nano	0.08
Claude Haiku 4.5	0
GPT-5.4 mini	0

Table 2

Task 2 Pass rate	
Claude Opus 4.6	0.84
Gemini 3 Flash Preview	0.76
Claude Sonnet 4.6	0.72
Gemini 3.1 Pro Preview	0.68
Gemini 2.5 Flash	0.24
GPT-5.4	0.08
Claude Haiku 4.5	0.04
GPT-5.4 nano	0.04
GPT-5.4 mini	0

Table 3

Task 3 Pass rate	
Gemini 3 Flash Preview	0.96
Claude Opus 4.6	0.92
Gemini 3.1 Pro Preview	0.88
Claude Sonnet 4.6	0.84
Gemini 2.5 Flash	0.76
GPT-5.4	0.48
Claude Haiku 4.5	0.04
GPT-5.4 nano	0.04
GPT-5.4 mini	0

Table 4

Model	Overall Runtime
Claude Opus 4.6	35m 40s
Claude Sonnet 4.6	37m 48s
Claude Haiku 4.5	14m 34s
GPT 5.4	8m 36s
GPT 5.4 mini	6m 44s
GPT 5.4 nano	6m 43s
Gemini 3.1 pro preview	61m 19s
Gemini 3 flash preview	97m 7s
Gemini 2.5 flash	72m 35s

Table 5